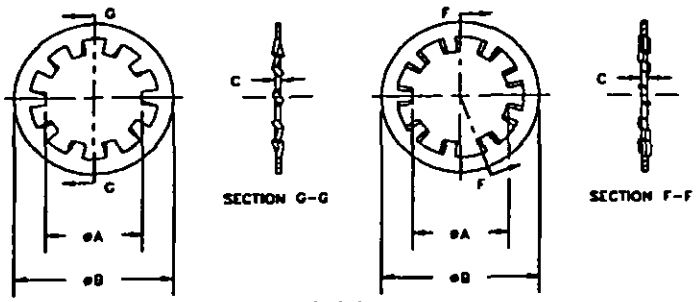
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ACCEPTABLE DESIGNS

TABLE I

NOMINAL SIZE	øA INSIDE DIAMETER		øB OUTSIDE DIAMETER		C THICKNESS		STEEL			TIN-BRASS OR PHOSPHOR BRONZE
							CARBON CAD OR ZINC (K)	CORROSION RES (K) COMP 410	CORROSION RES 300 SER.	
	MAX	MIN	MAX	MIN	MAX	MIN	DASH NO.	DASH NO.	DASH NO.	DASH NO.
#2 .086	.095	.089	.200	.175	.015	.010	35	69		103
#4 .112	.123	.115	.270	.255	.019	.015	36	70	153	104
#6 .138	.150	.141	.295	.275	.021	.017	37	71	154	105
#8 .164	.176	.168	.340	.325	.023	.018	38	72	155	106
#10 .190	.204	.195	.381	.365	.025	.020	39	73		107
1/4 .250	.267	.258	.478	.460	.028	.023	40	74		108
5/16 .312	.332	.320	.610	.594	.034	.028	41	75		109
3/8 .375	.398	.384	.692	.670	.040	.032	42	76		110
7/16 .438	.464	.448	.789	.740	.040	.032	43	77		111
1/2 .500	.530	.512	.900	.887	.045	.037	44	78		113
5/8 .625	.683	.640	1.071	1.045	.050	.042	46	80		115
3/4 .750	.795	.769	1.245	1.220	.055	.047	47	81		116
7/8 .875	.927	.894	1.410	1.364	.060	.052	48	82		117
1 1.000	1.060	1.019	1.637	1.550	.067	.059	49	83		118
1-1/4 1.250	1.325	1.275	1.975	1.921	.067	.059	51	85		120

TABLE II

NOMINAL SIZE	øA INSIDE DIAMETER		øB OUTSIDE DIAMETER		C THICKNESS		STEEL			TIN-BRASS OR PHOSPHOR BRONZE
							# CARBON CAD OR ZINC (K)	CORROSION RES (K) COMP 410	CORROSION RES 300 SER.	
	MAX	MIN	MAX	MIN	MAX	MIN	DASH NO.	DASH NO.	DASH NO.	DASH NO.
1/4 .250	.267	.256	.410	.395	.020	.016	125	135	187	145
5/8 .375	.392	.383	.507	.493	.030	.020		134		
7/16 .437	.448	.440	.592	.580	.020	.016	131	141		151
15/32 .468	.480	.472	.607	.593	.023	.013	126	136		146
9/16 .563	.580	.570	.690	.680	.023	.013	127	137		147
5/8 .625	.640	.630	.800	.790	.027	.017	128	138		148
3/4 .750	.785	.775	1.077	1.047	.027	.017	129	139		149
1 1.000	1.020	1.010	1.390	1.360	.028	.023	130	140		150

* Washers shown in Table II are generally to be considered for use in mounting electronic components
For carbon steel washers with black finish requirement add "B" after dash number. See requirement 2.1 and note 4.

(K) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY: DLA-IS CUSTODIANS: ARMY- AR NAVY- AS AIR FORCE- 99 DLA- REVIEW: AT, AV, ER, ES, GL, MI, NS, YD, 52 USER: MC, ME, OS, YD PROJECT NUMBER: 5310-1901	MILITARY SPECIFICATION SHEET TITLE WASHER, LOCK, FLAT-INTERNAL TOOTH	SPECIFICATION SHEET NUMBER MS35333 13 DEC 93 REV K SUPERSEDING MS35333 J 9 FEB 81 (SEE NOTE 4.) AMSC- N/A FSC 5310
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In a direction opposite to that of the other teeth.

REQUIREMENTS:

(K) 1. MATERIAL: Carbon, Steel, Grade 1050 to 1084, hardness40 to 50 HRC or equivalent.
Corrosion-resistant Steel, Composition 410.
Corrosion-resistant Steel, 300 series (UNS S300XX), hardness 35-43HRc.
Phosphor Bronze, Composition A in accordance with QQ-P-750..
Tin-brass, CDA Alloy 425 (UNS CA42500).

(K) 2. PROTECTIVE COATING: Carbon steel - Cadmium plated in accordance with QQ-P-416 type II, class 2, or Mechanically deposited Zinc in accordance with MIL-C-81562, Type II, Class 6.
Corrosion-resistant steel - Passivate in accordance with QQ-P-35.

(K) 2.1 BLACK PROTECTIVE COATING: When specified, Carbon steel shall be black (Electrodeposited) Chromium (olc) plating in accordance with MIL-C-14538.

(K) 3. PART NUMBER: The MS part number consists of the MS number, plus the dash number, plus the code letter, when applicable.

CODE LETTERS:
Add "B" as suffix to dash number for black finish per requirement 2.1.
Example: MS35333-131B Washer, Carbon Steel, Black Finish.

(K) 4. PACKAGING: Packaging shall be in accordance with ASTM D3951.

NOTES:
1. All dimensions are in inches.
2. The number of teeth shall be at the option of the Supplier. (K)
3. Washer, dash numbers 121 and 112 only, have the alternate teeth twisted
4. MS35333 supersedes AN936 in part, BEAX 1 and BEAX 2.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of the standard shall take precedence.
6. Referenced government (or non-government) documents of the issue in that issue of the Department of Defense Index of the Specification and Standards (DoDISS) specified in the solicitation form a part of the standard to the extent specified herein.

INTERCHANGEABILITY TABLE

The following dash numbers of MS35333 and AN936 are cancelled after 5 Dec 1963. The cancelled washers cannot always replace the new washers and should be used until existing stocks are depleted. Use only the new washers for design and engineering. Replacement shall be in accordance with this table.

TABLE III

PART NUMBERS											
CANCELLED		NEW	CANCELLED		NEW	CANCELLED		NEW	CANCELLED		NEW
AN936	MS35333	MS35333	AN936	MS35333	MS35333	AN936	MS35333	MS35333	AN936	MS35333	MS35333
A2	1	35	--	57	40	A1016	12	46		86	103
--	18	35	A416H	--	**	--	29	46	A4B	87	104
--	52	35	A516	7	41	--	63	46	A6B	88	105
A3	--	**	--	24	41	A1215	13	47	A8B	89	106
A4	2	36	--	58	41	--	30	47	A10B	90	107
--	19	36	A616	8	42	--	64	47	A416B	91	108
--	53	36	--	25	42	A1416	14	48	A516B	92	109
A6	3	37	--	59	42	--	31	48	A616B	93	110
--	20	37	A716	9	43	--	65	48	A716B	94	111
--	54	37	--	26	43	A1616	15	49	A716BS	--	112
A8	4	38	--	60	43		32	49		95	113
--	21	38	A716S		121		66	49		96	114
--	55	38	A816	10	44		16	50		97	115
A10	5	39	--	27	44		33	50		98	116
--	22	39	--	61	44		67	50		99	117
--	56	39	A916	11	45		17	51		100	118
A416	6	40	--	28	45		34	51		101	119
--	23	40	--	62	45		68	51		102	120

**No replacement

PREPARING ACTIVITY: DLA-IS CUSTODIANS: ARMY- AR NAVY- AS AIR FORCE- 99 DLA- REVIEW: AT, AV, ER, ES, GL, MI, NS, YD, 82 USER: MC, ME, OS, YD PROJECT NUMBER: 5310-1901	MILITARY SPECIFICATION SHEET TITLE WASHER, LOCK, FLAT-INTERNAL TOOTH	SPECIFICATION SHEET NUMBER MS35333 13 DEC 93 REV K SUPERSEDING MS35333 J 9 FEB 81 (SEE NOTE 4.) AMSC- N/A FSC 5310	
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